
DTU

02159 – Operating Systems

Week 1: Introduction to 02159 and 02335

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Do you know any Operating Systems?

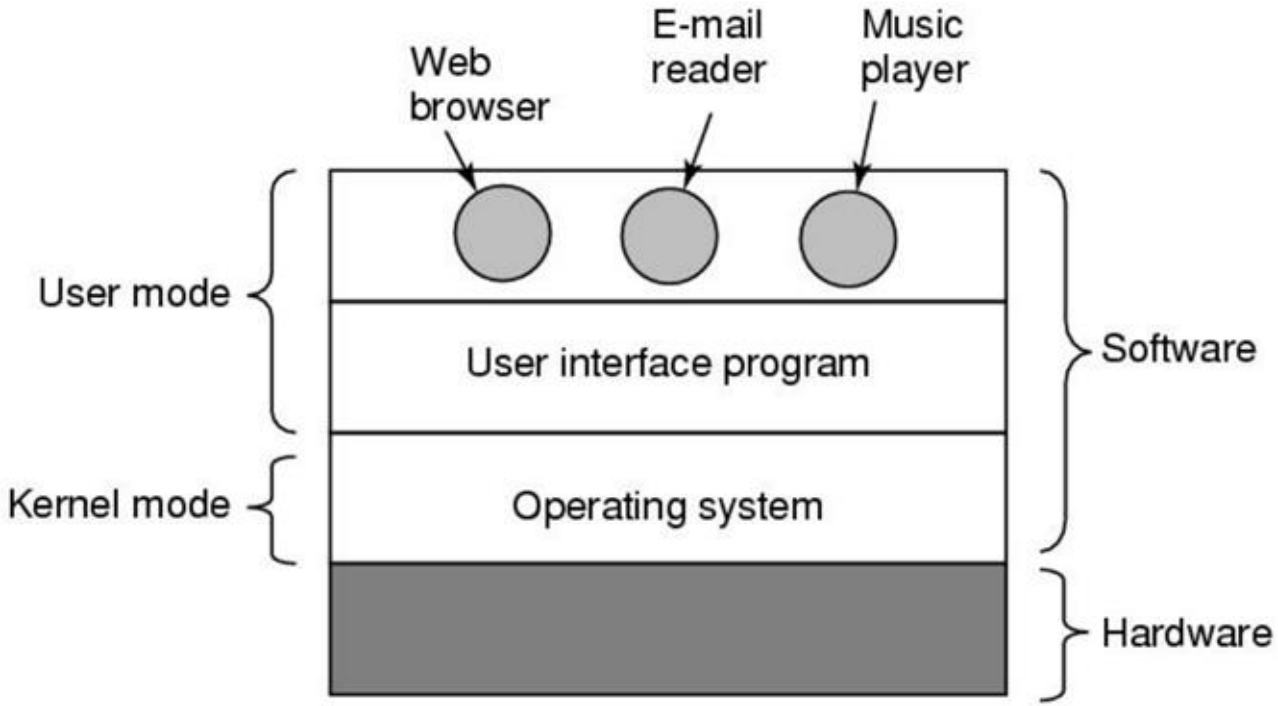
Operating Systems

- Desktops, Laptops, Servers
  - Windows, macOS, Linux, Unix, FreeBSD, OpenBSD
- Smartphones, Smart Watches, Smart TVs
  - Android, iOS, Blackberry OS, Symbian
- Networking Devices, Routers
  - Cisco IOS, OpenWrt
- Embedded Devices (The things of the Internet of Things)
  - Contiki OS, RIOT, OpenWSN, TinyOS

What do Operating Systems do?

Operating Systems

- Operating Systems **manage** and **abstract** resources



Operating Systems

- Are there any computer systems that do **not** use an operating system?

Operating Systems

- Are there any computer systems that do **not** use an operating system?
  - Bare metal / bare machine

Practical Information about the course

02159 and 02335

- 02159 and 02335 will run as the same course in 2026
  - Same classroom
  - Same lectures
  - Same project

Some Statistics from 2018

- Average grade of students that did **all** the hands-on assignments: **10.2/12**
- Average grade of students that did **only the mandatory** hands-on assignments: **6.6/12**
- Hands-on assignments facilitate deep learning

OS Challenge

- Idea: **Let's make it fun!**
- Convert the hands-on programming assignments into a **game**
- Learn OS by playing the game
- The team that provides the best solution to the challenge wins

OS Challenge: Inspired by PBL

- The OS challenge is inspired by PBL (Problem-Based Learning)
  - Students form groups and face a problem
  - Students break the problem into sub-problems, identify missing knowledge and specify their own individual learning objectives
  - Teacher facilitates the process but provides no solution to the problem
  - Supplementary lectures assist students towards the solution

Course Structure: Learning Activities

- **Lectures** (video lectures also provided)
  - The lectures will discuss various OS concepts
  - And inspire ideas on how to address the OS Challenge
  - Supplementary video lectures
  - Supplementary **self-study lectures** on C Programming and TCP/IP Networking

- The **OS Challenge** will be the course project
  - No lectures on some weeks to focus on project with TA support

Teaching Staff

- Main Teacher
  - Xenofon (Fontas) Fafoutis xefa@dtu.dk
- Co-Teachers
  - Charalampos (Haris) Orfanidis chaorf@dtu.dk
  - Roberto Morabito romor@dtu.dk
- Teaching Assistants
  - Contact details on DTU Learn

Where, When, What

- Homework prior to lecture (anywhere)
  - Watch weekly video lecture, study theory or catch up with project
- Lecture Days
  - Lecture
  - Group meeting (discuss lecture/project), project work
- Project Days
  - Group meeting (discuss lecture/project), project work
  - Get support from TAs

Xenofon (Fontas) Fafoutis

- Originally from Greece, I got my PhD from DTU Compute in 2014
- Researcher at Bristol, UK, for over 4 years
- Faculty at DTU Compute (since Aug 2018)
  - Embedded Systems Engineering section
- Research Interests
  - Wireless Embedded Systems, Embedded AI, Low-Power Networks, Internet of Things
- Interested in final year projects?
  - Check: www.compute.dtu.dk/~xefa

Communication

- Announcements on DTU Learn (check your emails!)
- Course website: DTU Learn
- Email: xefa@dtu.dk
- Office: 322/120 (by appointment only)

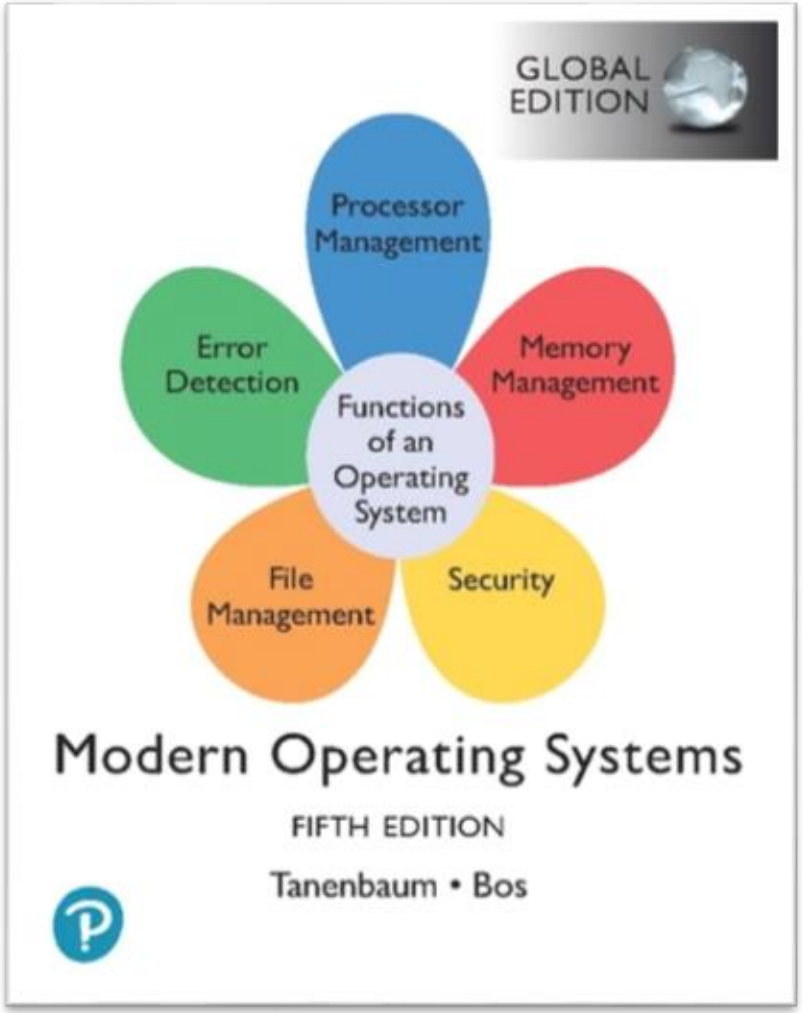
Language

- English (only)

Reading Material

- Textbook
  - A. S. Tanenbaum and H. Bos, **Modern Operating Systems**, 5th edition. Pearson Education Inc., 2023.

- Additional material will be linked on DTU Learn



Assessment

- Overall Assessment
  - Mandatory group project + final written examination
- OS Challenge
  - Based on how well you meet the learning objectives with objective criteria
  - Not on the rankings
- Final Written Examination
  - All aids are allowed